Quasi-steady-state lap time simulation

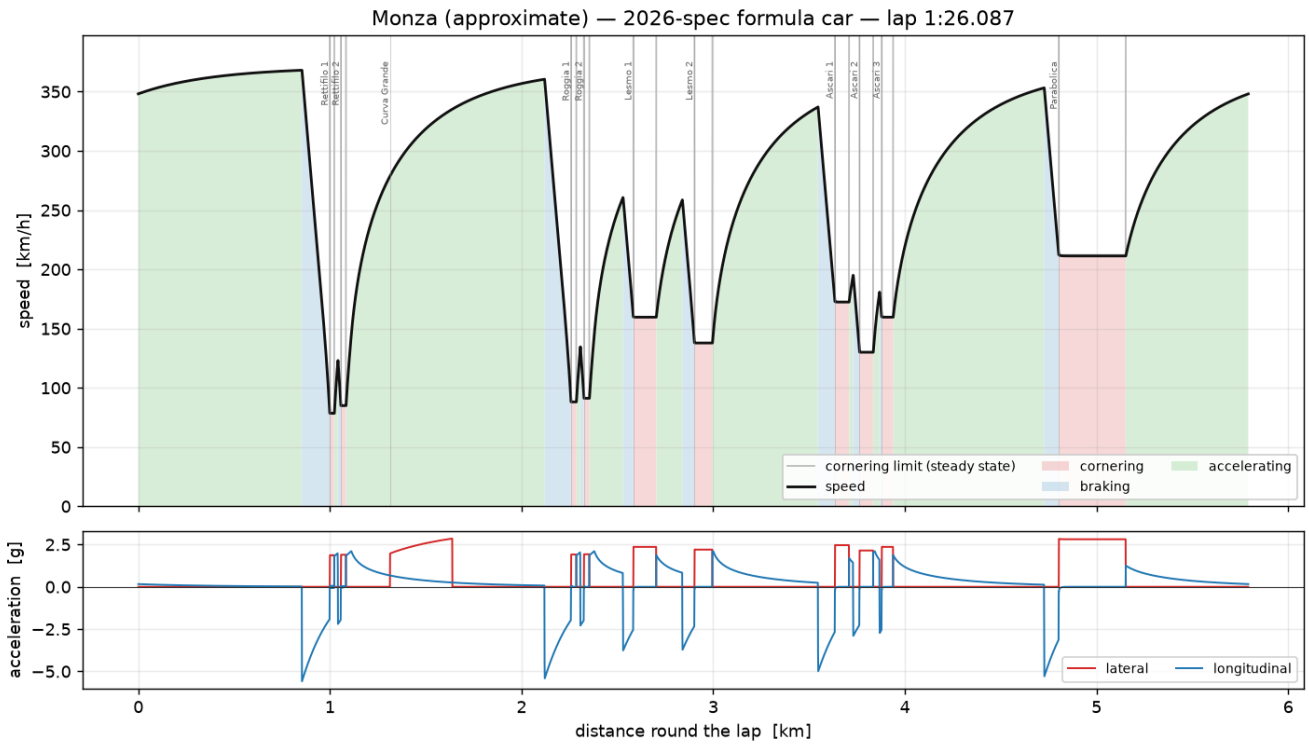
2026-spec formula car — 800 kg, 750 kW, CLA 3.60 m², CDA 1.05 m², L/D 3.43, μ 1.75 with load exponent 0.25

3 circuits, agreeing with reality to within 11.6%

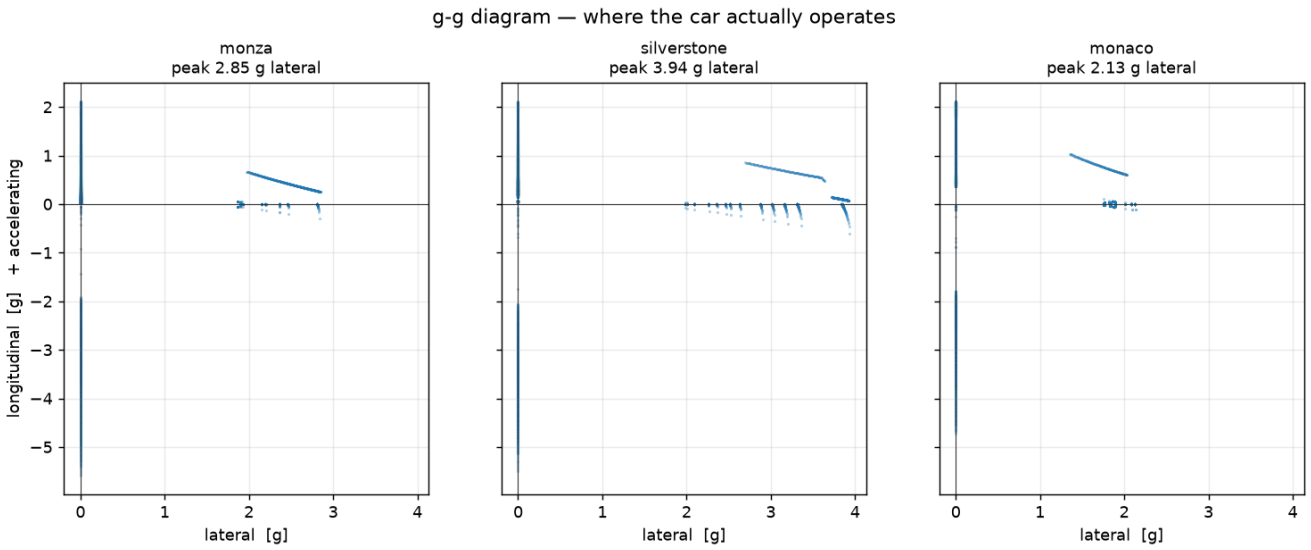
Best agreement +3.1% at monza, worst +11.6% at monaco. The irreducible RMS after calibration is 4.81 s.

3	+3.1%	+11.6%	4.81 s
circuits simulated	best agreement	worst agreement	irreducible RMS

Read the absolute lap times with care. The circuit geometry is estimated from published layouts, not surveyed, and every corner is a constant-radius arc. The sensitivities and the downforce trade below are far more robust than any single lap time, because the geometry errors largely cancel when the same track is run twice with different cars.



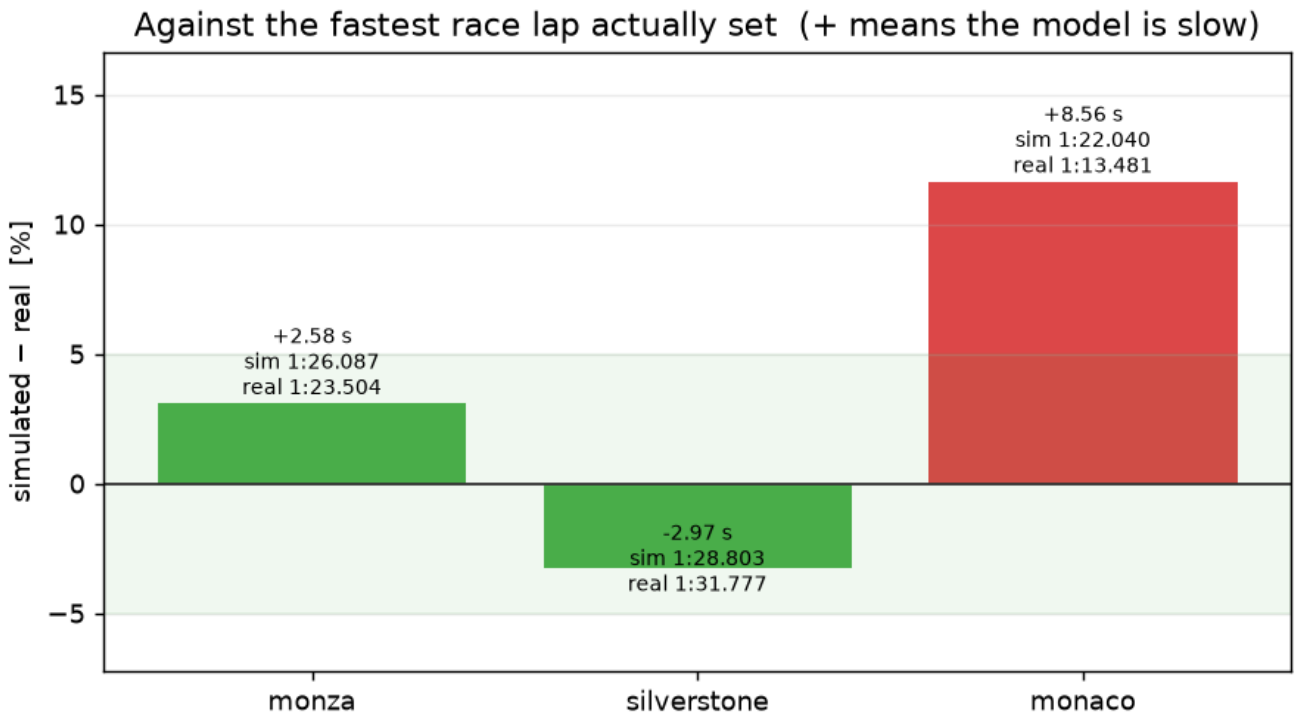
Simulated laps

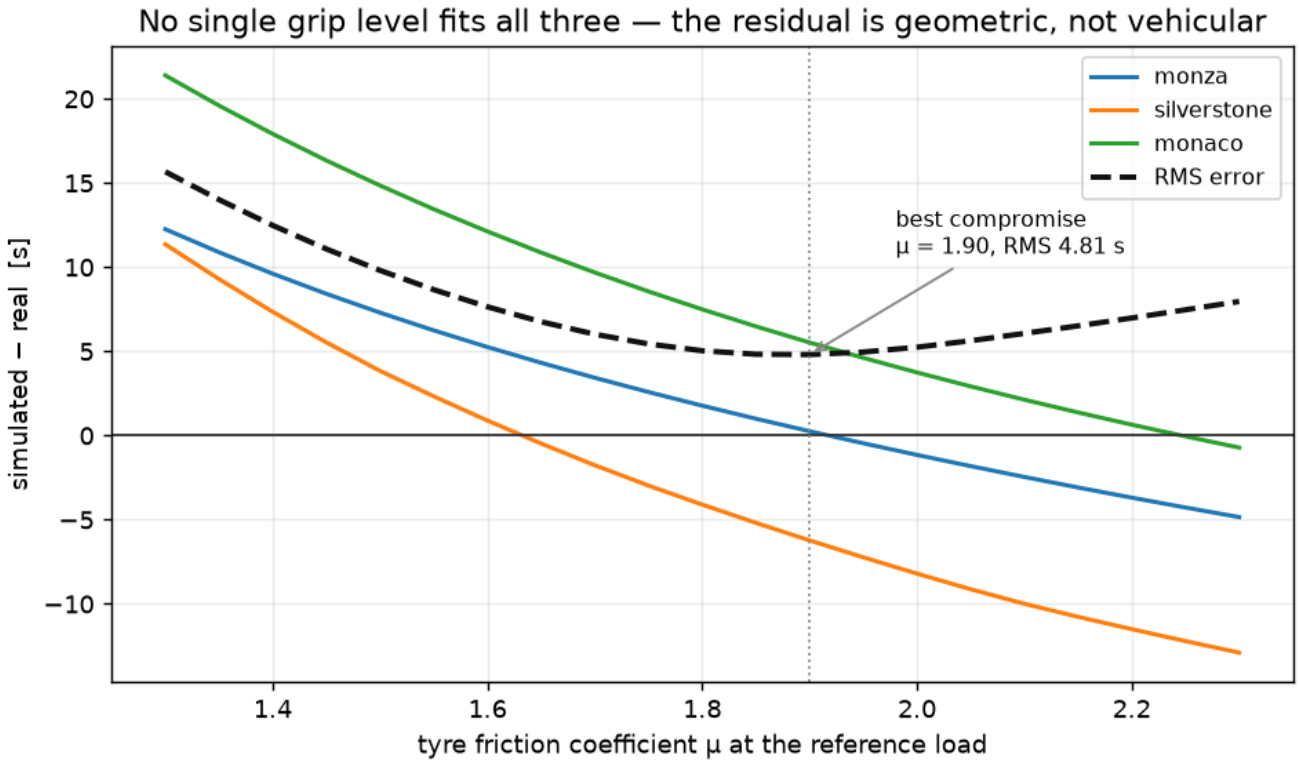


circuit	lap	top km/h	min km/h	mean km/h	peak lat g	full throttle
monza	1:26.087	368.1	78.5	242.2	2.85	62.7%
silverstone	1:28.803	365.1	102.9	238.8	3.94	53.4%
monaco	1:22.040	320.8	44.7	146.4	2.13	42.0%

Against reality

The reference is the fastest race lap actually set at each circuit, which is a lower bound on what the car could do there, not a target.



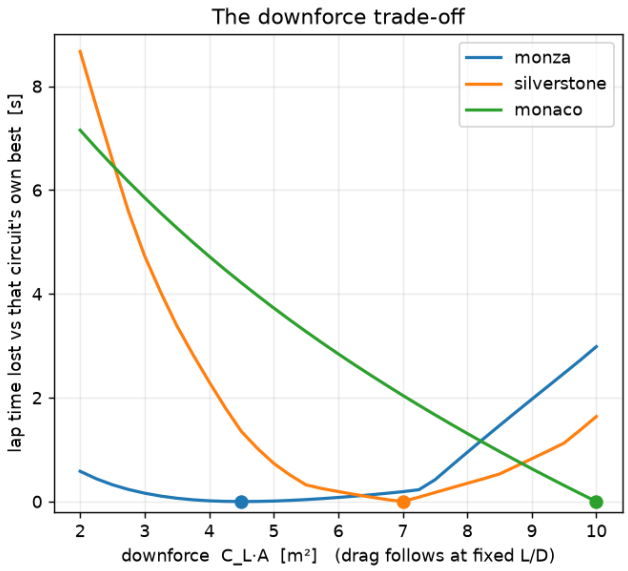


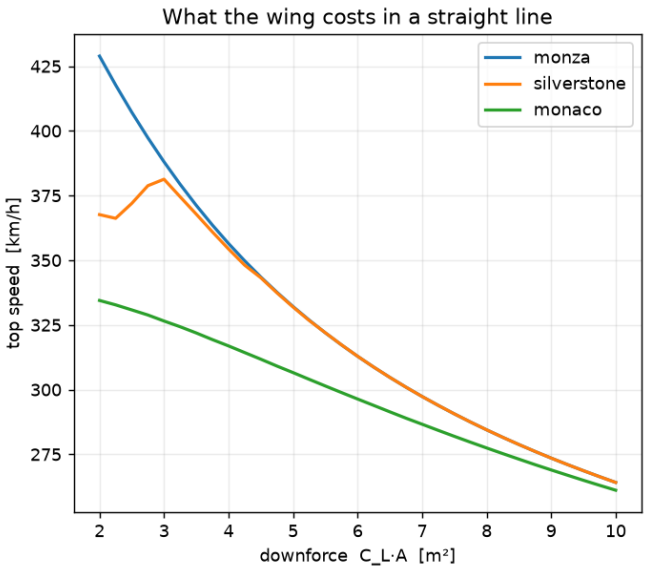
Against reality (continued)

circuit	simulated	fastest race lap	driver	error	error
monza	1:26.087	1:23.504	Antonelli	+2.58 s	+3.1%
silverstone	1:28.803	1:31.777	Antonelli	-2.97 s	-3.2%
monaco	1:22.040	1:13.481	Antonelli	+8.56 s	+11.6%

The downforce trade-off

The optimum is flat: the "within 0.1 s" column is the range of CLA that costs less than a tenth, and it is wide enough at most circuits that the setup choice is being made on something other than lap time.





circuit	optimum CLA	lap	top km/h	within 0.1 s
monza	4.50	1:26.037	343.5	3.50 – 6.00
silverstone	7.00	1:25.652	297.3	6.50 – 7.25
monaco	10.00	1:16.882	261.1	10.00 – 10.00

Taking the wrong wing to the wrong circuit

Each row is one setup run at every circuit. The number after each lap is what that setup loses against the best setup for that circuit.

	monza	silverstone	monaco
monza setup	1:26.037 best	1:27.001 +1.35	1:21.094 +4.21
silverstone setup	1:26.227 +0.19	1:25.652 best	1:18.927 +2.04
monaco setup	1:29.021 +2.98	1:27.290 +1.64	1:16.882 best

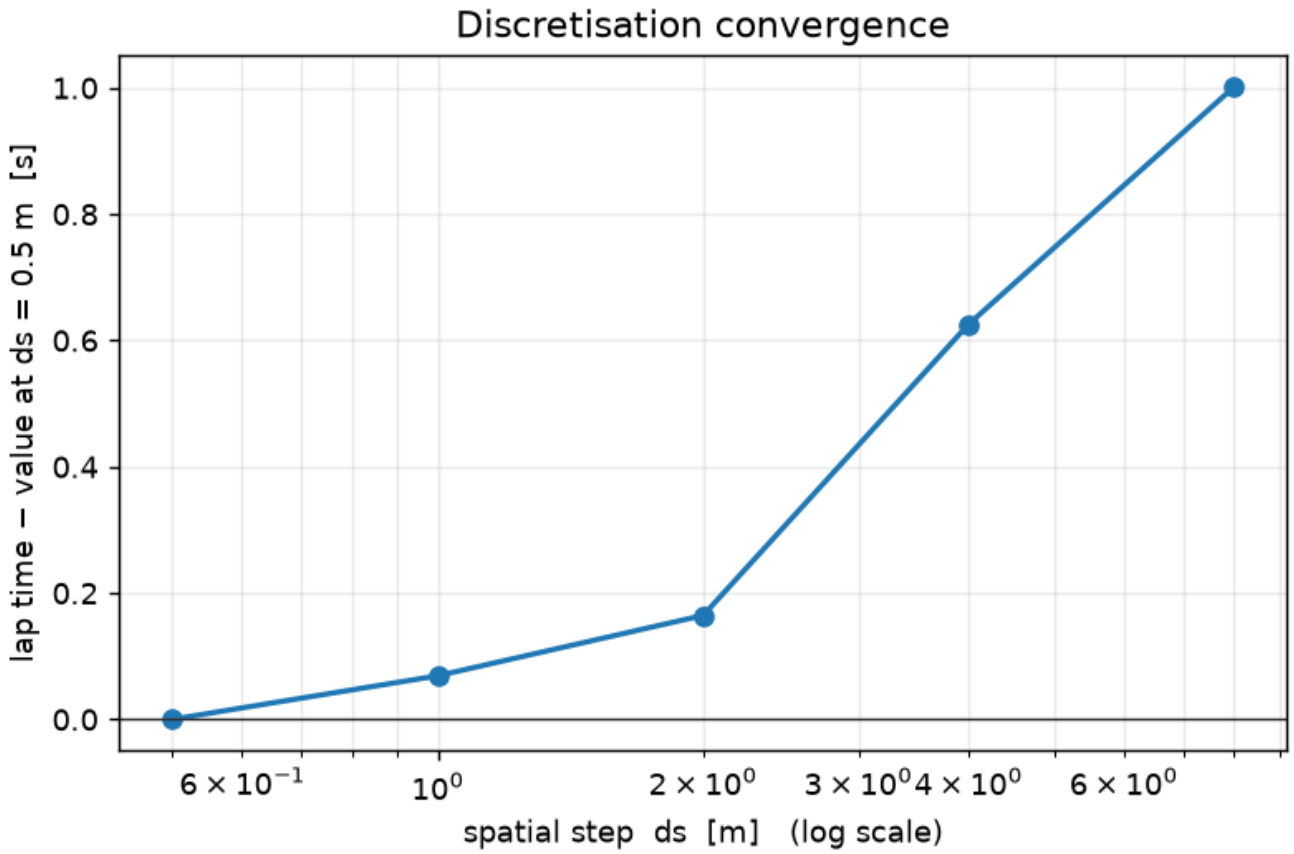
Sensitivities

What one step of each parameter is worth at the reference circuit, holding everything else fixed.

parameter	per	lap time	of a lap
mass	10 kg	+0.275 s	+0.32%
power	10 kW	-0.190 s	-0.22%
mu	0.1 grip	-1.657 s	-1.92%
cda	0.1 m2 drag	+0.644 s	+0.75%
cla	0.5 m2 downforce	-1.001 s	-1.16%

Numerics

The solver is run at several step sizes so the discretisation error can be seen rather than assumed. The lap time quoted above is from the finest grid.



Seeded and deterministic: re-running reproduces every number above.